Software Requirement Specification Report

For

Study Abroad Preparation System

Course Title: Software Engineering Design Capstone Project

Course Code: SE 331

Submitted To

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Software Requirements Specification (SRS) for StudyGlobe

1. Introduction

1.1 Problem Statement

Students planning to study abroad often face difficulties in collecting and organizing information related to universities, programs, admission requirements, tuition fees, and scholarships. Currently, this information is scattered across multiple websites and sources, making the preparation process time-consuming and confusing. Many students rely on manual searching, spreadsheets, or informal guidance, which may lead to missed opportunities or incorrect decisions. There is a need for a centralized platform that organizes study abroad preparation information in a structured and accessible manner.

1.2 Purpose

The purpose of the Study Abroad Preparation System is to provide a web-based platform that helps students explore and organize essential information related to studying abroad. The system aims to:

- Provide structured access to country-wise universities and programs
- Present admission requirements, intake schedules, and language criteria
- Display scholarship eligibility information
- Allow students to filter options based on preferences
- Reduce the effort required to gather and compare information manually

1.3 Scope

The system focuses on the information gathering and preparation phase of studying abroad. It allows users to explore universities, programs, requirements, and eligibility criteria through a centralized platform.

In-Scope Features:

- Provide browsing and filtering of universities and programs
- Display intake schedules, language requirements, and tuition information
- Show scholarship eligibility criteria
- Allow profile-based preference filtering

- Provide administrative management of stored data

Out-of-scope Features:

- Process university applications
- Handle visa procedures
- Guarantee admission or scholarship outcomes
- Replace official university sources

2. Design and Implementation Constraints

2.1 Programming Language

The system will use a web-based technology stack:

- Language: JavaScript
- Frontend: React with Tailwind CSS for responsive interface design
- Backend: Node.js for API development & Python for web scraper.

This stack is chosen because it supports rapid development, scalability, and seamless integration between frontend and backend components.

2.2 Database & Servers

- **Database:** PostgreSQL (via Supabase)
- **Purpose:** Store structured data including countries, universities, programs, intakes, language requirements, and scholarship eligibility
- **Communication:** RESTful APIs will be used to retrieve and manage data between the frontend and backend

The relational database model ensures data consistency and supports efficient filtering and querying operations.

3. User Classes and Characteristics

Students (Primary Users)

- Explore countries, universities, and programs
- View requirements and eligibility information
- Manage personal preferences
- Use filtering features to narrow options

Students are assumed to have basic familiarity with web applications.

System Administrator

- Manage the database content
- Add, update, or remove countries, universities, programs, and requirements
- Ensure data accuracy and consistency

Administrators are expected to have technical familiarity with the system.

Additionally, academic instructor provides project requirements and evaluation but does not interact directly with the system.

4. Functional Requirements (FR)

ID	Requirement Name	Description	Users	Priority
FR-01	Secure Enrollment and Authentication	The system shall allow users to register and log in securely.	Students	High
FR-02	User Profile Management	The system shall allow users to create and update their academic details and preferences.	Students	High
FR-03	Country Catalog	The system shall display a list of available countries for study abroad.	Students	High
FR-04	View Country-wise Universities	The system shall allow users to view universities based on selected country.	Students	High
FR-05	University Details	The system shall display detailed information about a selected university.	Students	High
FR-06	Academic Program Exploration	The system shall allow users to view programs offered by a selected university.	Students	High

FR-07	View Intake Schedule	The system shall display available intake sessions for a university.	Students	Medium
FR-08	Language Requirement Inspection	The system shall display language test requirements for programs.	Students	Medium
FR-09	Scholarship Eligibility Assessment	The system shall display scholarship eligibility information for universities.	Students	Medium
FR-10	Admin Data Management	The system shall allow administrators to manage stored data.	System Admin	High
FR-11	Preference-Based Filtering	The system shall allow users to filter universities based on preferences.	Students	High

5. Non-Functional Requirements

Performance

- The system should respond to user requests within 3 seconds under normal usage conditions.

- Filtering and data retrieval operations should complete without noticeable delay.

- The system should support multiple users accessing information simultaneously without performance degradation.

Security

- User authentication must be required to access personalized features.

- Passwords must be stored in encrypted form.

- Administrative functions must be restricted to authorized users only.

- The system should prevent unauthorized modification of stored data.

Usability

- The interface should be simple and intuitive for students with basic web browsing experience.

- Navigation between countries, universities, and programs should be clear and consistent.

- The system should be accessible through modern web browsers without requiring additional software.

Reliability

- The system should maintain data consistency across all modules.
- Stored information should remain available even after system restarts.
- Backup mechanisms should ensure minimal data loss in case of failure.

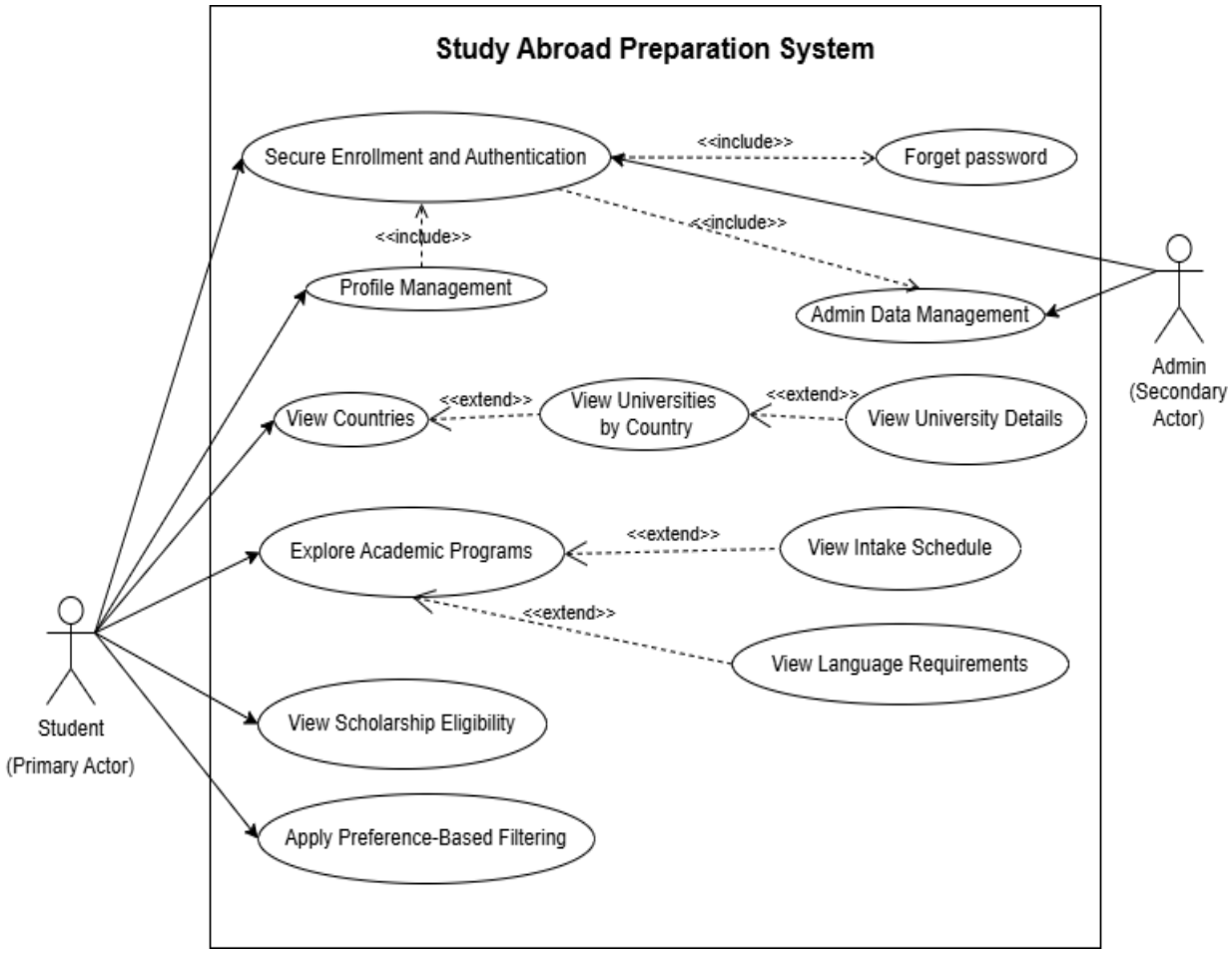
Scalability

- The system should allow addition of new countries, universities, and programs without requiring structural changes.
- The database design should support expansion of data volume over time.

Maintainability

- The system should allow administrators to update or correct data easily.
- Code and database structure should be organized to support future improvements.

6.1 Use Case Diagram:



6.2. Use Case Descriptions

UC-01: Secure Enrollment and Authentication

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. The user has internet access. 3. For login: the user must already have a registered account.
Main Success Flow	1. The user accesses the system homepage. 2. The user selects the option to register or log in. **Registration Flow (New User)** 3. The user enters required details (name, email, password). 4. The system validates the input data. 5. The system creates a new user account. 6. The system confirms successful registration. **Login Flow (Existing User)** 7. The user enters registered email and password. 8. The system verifies credentials. 9. The system grants access to the user dashboard.
Alternative Flows	1. Invalid Input During Registration : If required fields are missing or invalid, the system prompts the user to correct the input. 2. Incorrect Login Credentials : If the email or password is incorrect, the system displays an error message and allows retry.

UC-02: User Profile Management

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The user is logged into the system. 2. The system is operational.
Main Success Flow	1. The user navigates to the profile section. 2. The system displays the current profile information. 3. The user updates academic details and preferences (e.g., country, degree level, field, budget). 4. The user submits the updated information. 5. The system validates the input data. 6. The system saves the updated profile. 7. The system confirms successful update.
Alternative Flows	1. Invalid Input : If entered data is incomplete or invalid, the system prompts the user to correct the information. 2. Session Timeout : If the session expires, the system requires the user to log in again.

UC-03: Country Catalog

Primary Actor	Student
Secondary Actor	None
Preconditions	1.The system is operational.
Main Success Flow	1.The user accesses the country catalog section. 2.The system retrieves the list of available countries. 3.The system displays the countries to the user.
Alternative Flow	No Data Available : If no country data exists, the system displays a message indicating unavailability.

UC-04: View Country-wise Universities

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Country data is available in the system.
Main Success Flow	1. The user selects a country from the country catalog. 2. The system retrieves universities associated with the selected country. 3. The system displays the list of universities to the user.
Alternative Flow	No Universities Found : If no universities are available for the selected country, the system displays an appropriate message.

UC-05: University Details

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. University data exists in the system.
Main Success Flow	1. The user selects a university from the university list. 2. The system retrieves detailed information about the selected university. 3. The system displays the university details including overview and basic information.

Alternative Flow	Data Retrieval Error : If the system cannot retrieve university data, an error message is displayed.

UC-06: Academic Program Exploration

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Program data exists for the selected university.
Main Success Flow	1. The user selects a university. 2. The user navigates to the academic programs section. 3. The system retrieves programs offered by the selected university. 4. The system displays the list of programs with relevant details.
Alternative Flow	No Programs Available : If no programs are available, the system displays a notification.

UC-07: View Intake Schedule

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Intake data exists for the selected university.
Main Success Flow	1. The user selects a university. 2. The user navigates to the intake schedule section. 3. The system retrieves available intake sessions for the selected university. 4. The system displays the intake schedule to the user.
Alternative Flow	No Intake Information : If intake data is unavailable, the system displays an appropriate message.

UC-08: Language Requirement Inspection

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Language requirement data exists for the selected university.
Main Success Flow	1. The user selects a university. 2. The user navigates to the language requirements section. 3. The system retrieves language test requirements. 4. The system displays the required tests and minimum scores.
Alternative Flow	No Requirement Data : If language requirements are not available, the system displays a notification.

UC-09: Scholarship Eligibility Assessment

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Scholarship eligibility data exists for the selected university.
Main Success Flow	1. The user selects a university. 2. The user navigates to the scholarship eligibility section. 3. The system retrieves eligibility criteria for scholarships. 4. The system displays the eligibility information to the user.
Alternative Flow	No Scholarship Information : If no eligibility data is available, the system displays a message.

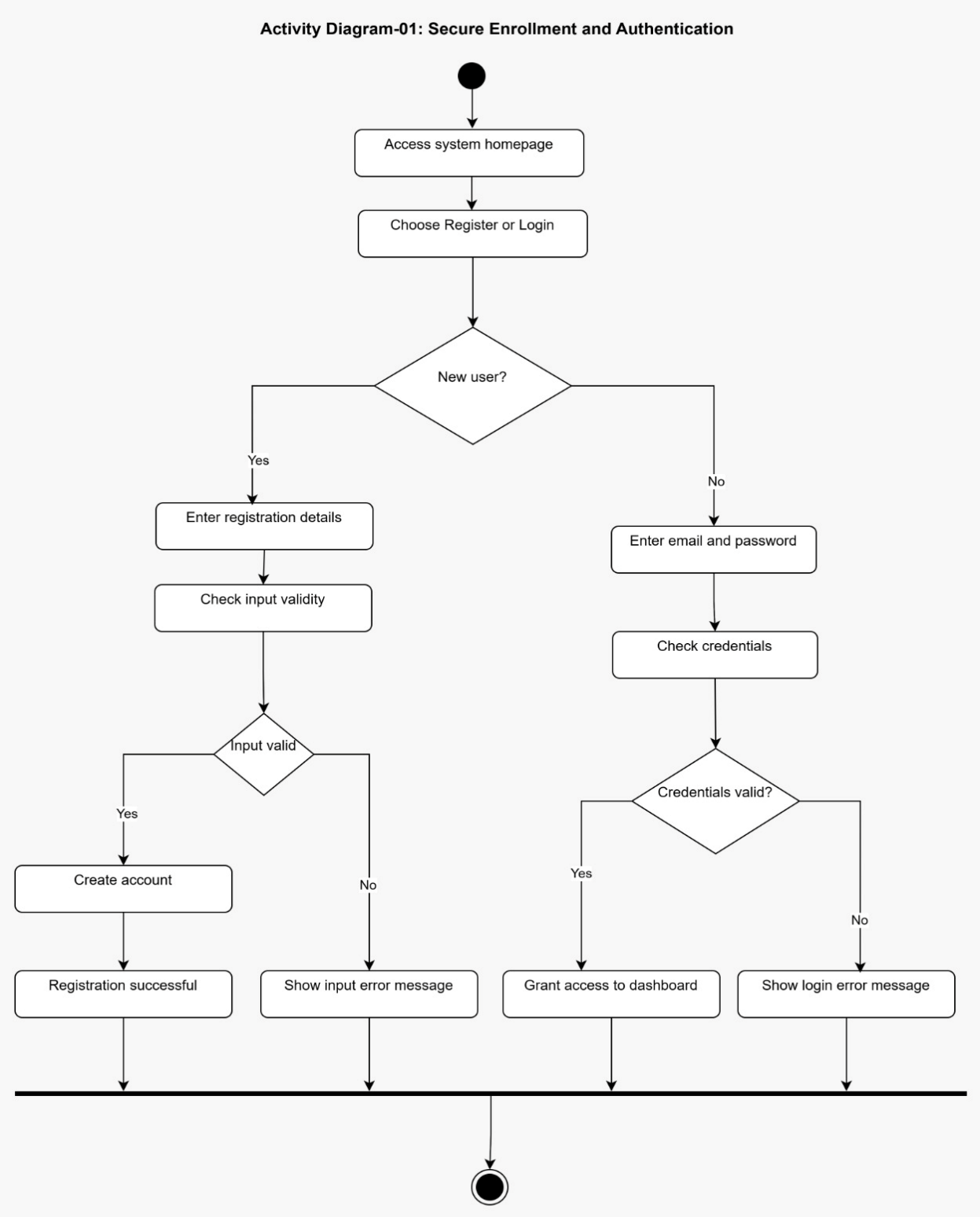
UC-10: Admin Data Management

Primary Actor	Admin
Secondary Actor	None
Preconditions	1. The administrator is logged into the system. 2. The system is operational.
Main Success Flow	1. The administrator accesses the admin panel. 2. The administrator selects a data category (country, university, program, intake, language requirement, or scholarship). 3. The system displays existing records for the selected category. 4. The administrator performs add, update, or delete operations. 5. The system validates the changes. 6. The system saves the updated data. 7. The system confirms successful data management operation.
Alternative Flows	1. Invalid Data Entry : If entered data is invalid, the system prompts correction. 2. Unauthorized Access : If a non-admin user attempts access, the system denies entry.

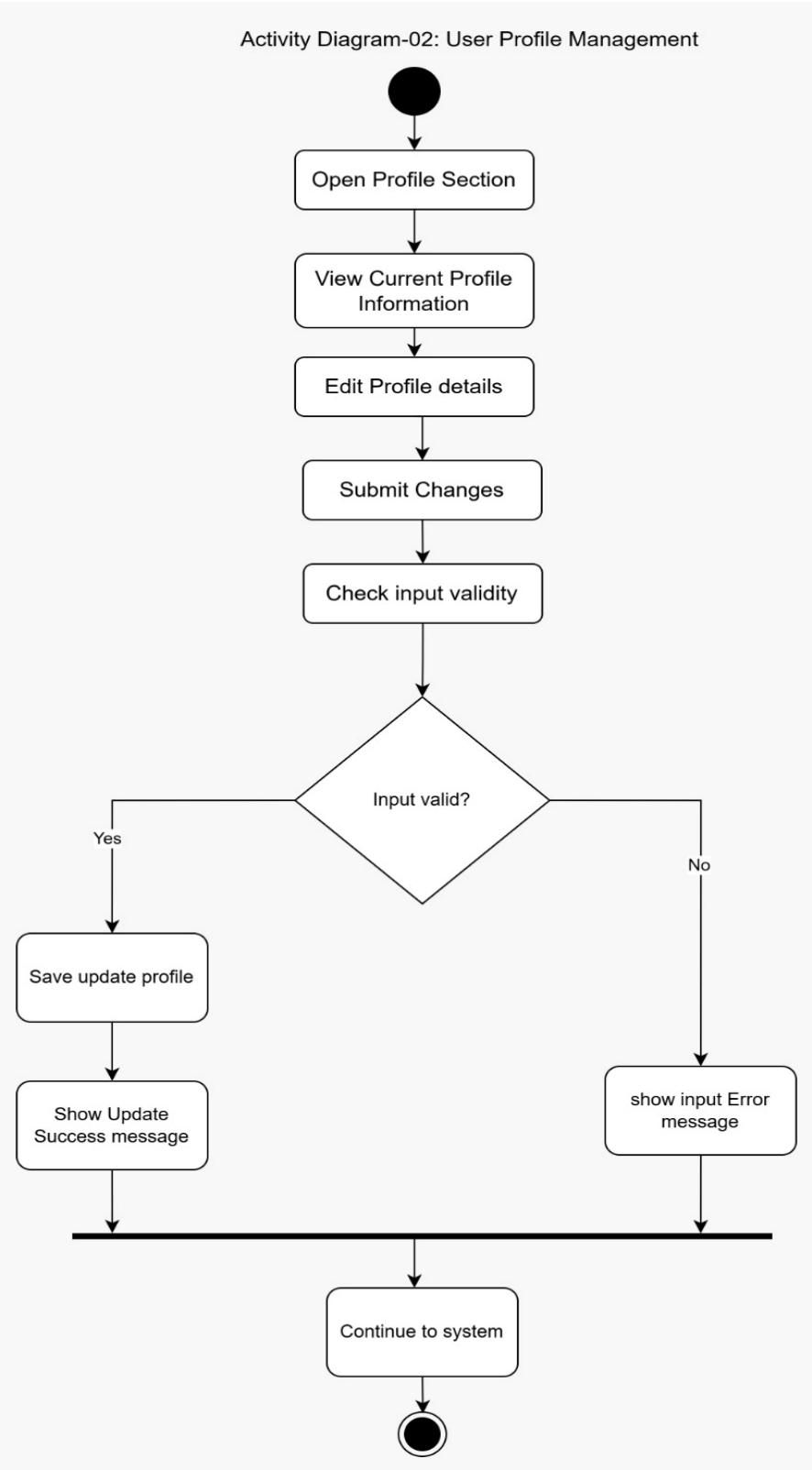
UC-11: Preference-Based Filtering

Primary Actor	Student
Secondary Actor	None
Preconditions	1. The system is operational. 2. Relevant university and program data exist.
Main Success Flow	1. The user navigates to the filtering section. 2. The user selects preferences such as degree level, field of study, or tuition range. 3. The user applies the filter criteria. 4. The system processes the selected preferences. 5. The system displays universities and programs matching the criteria.
Alternative Flow	No Matching Results : If no data matches the selected preferences, the system displays a message.

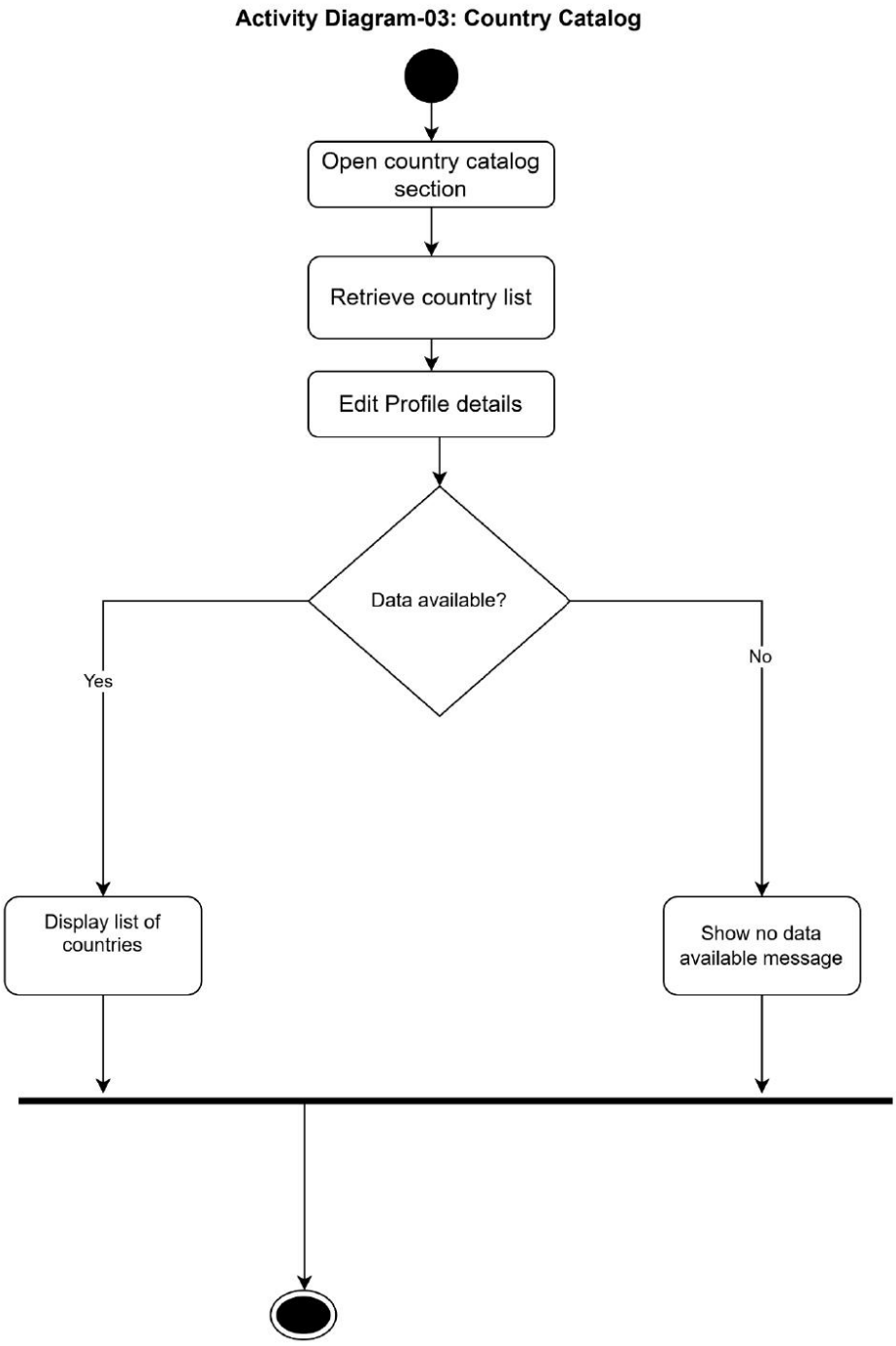
6.3 Activity Diagram



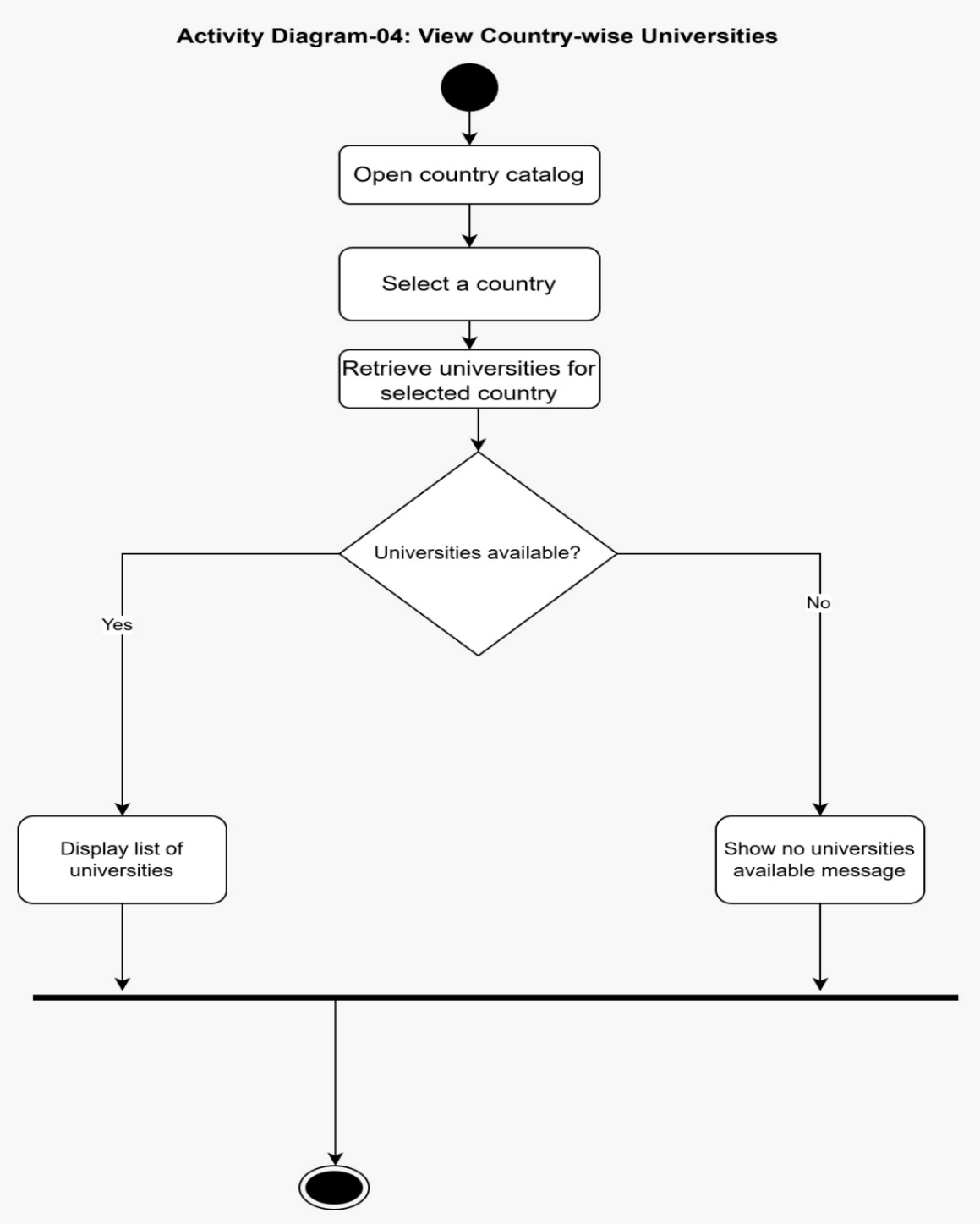
Activity Diagram-02: User Profile Management



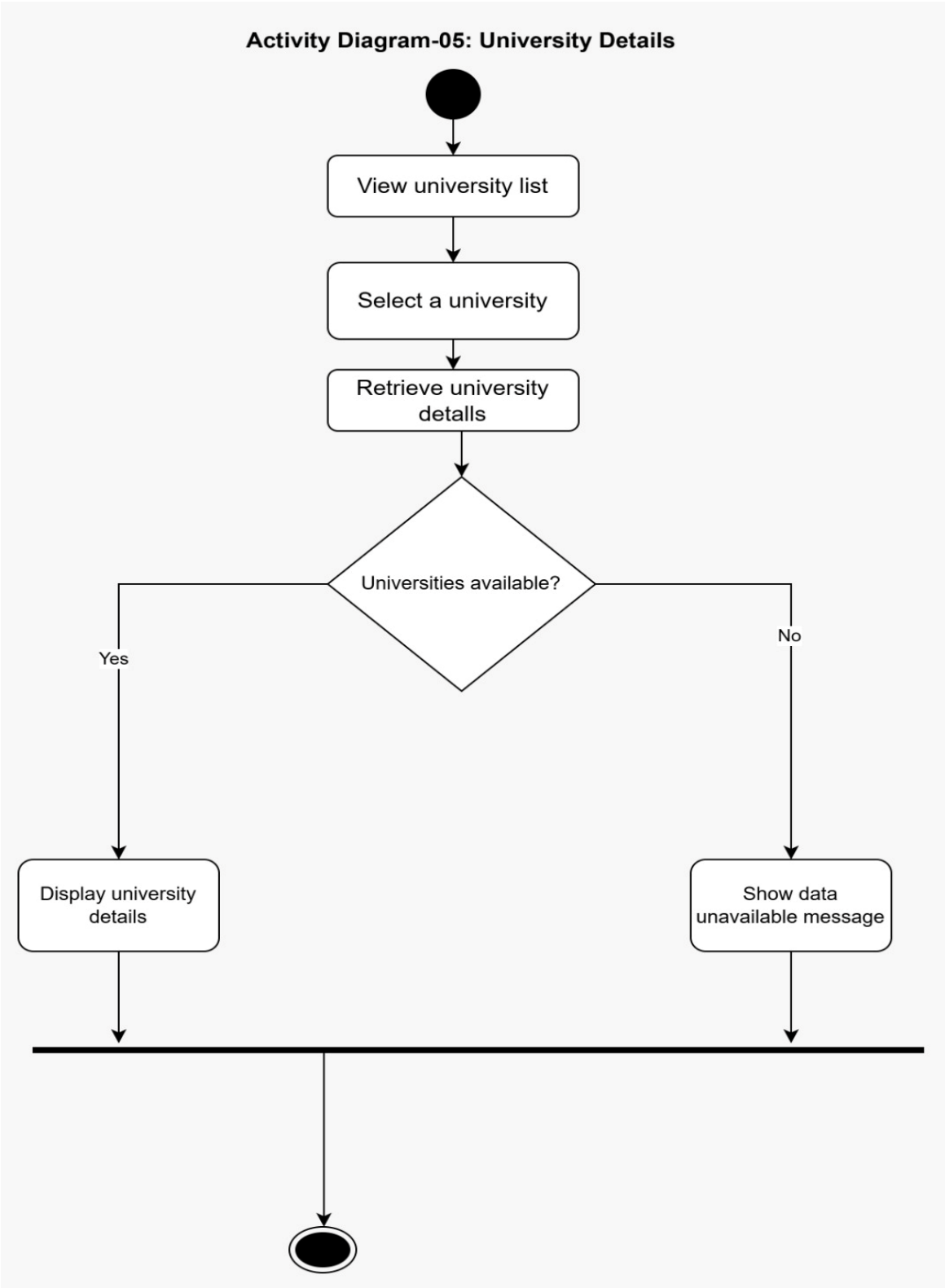
Activity Diagram-03: Country Catalog



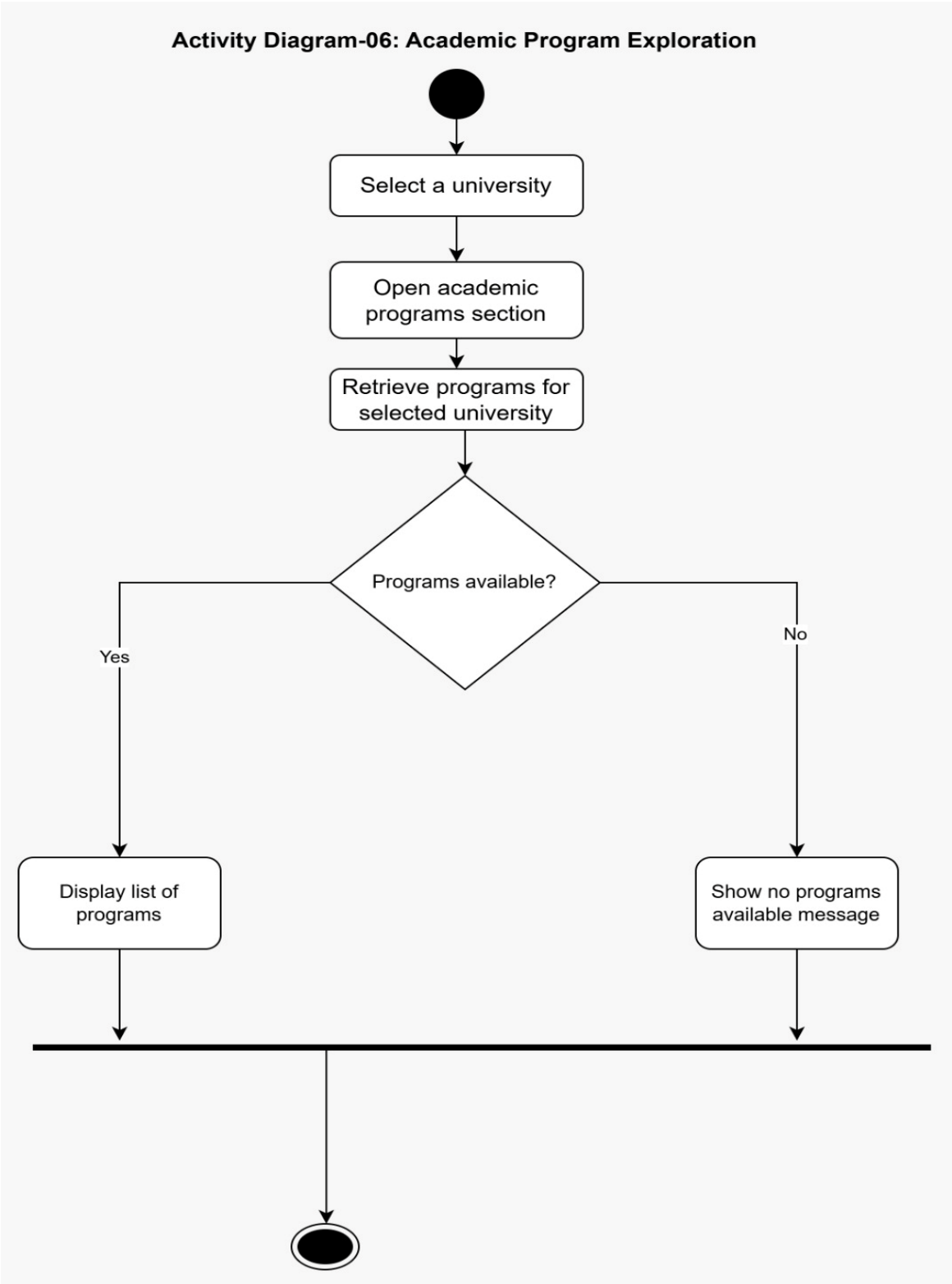
Activity Diagram-04: View Country-wise Universities



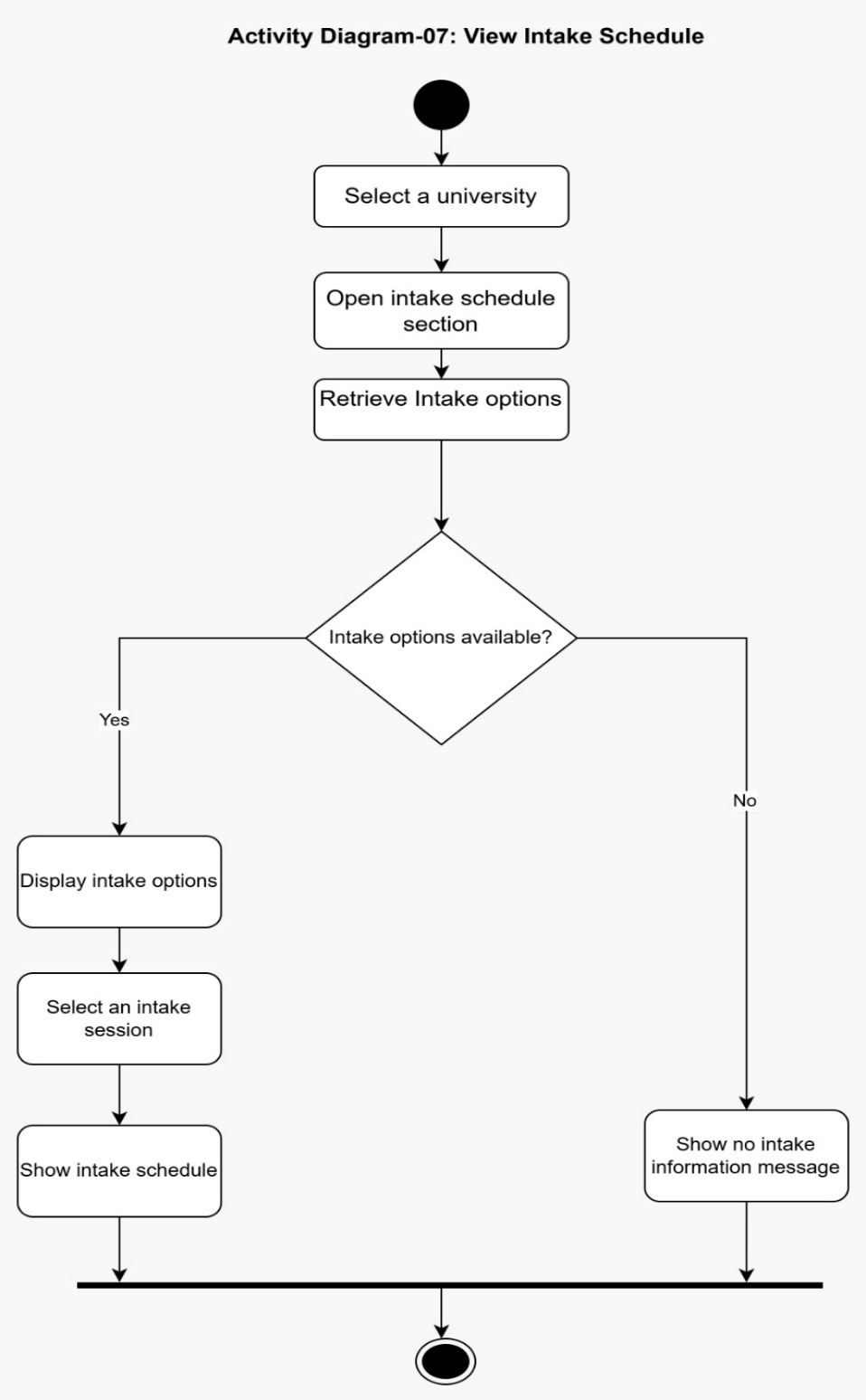
Activity Diagram-05: University Details



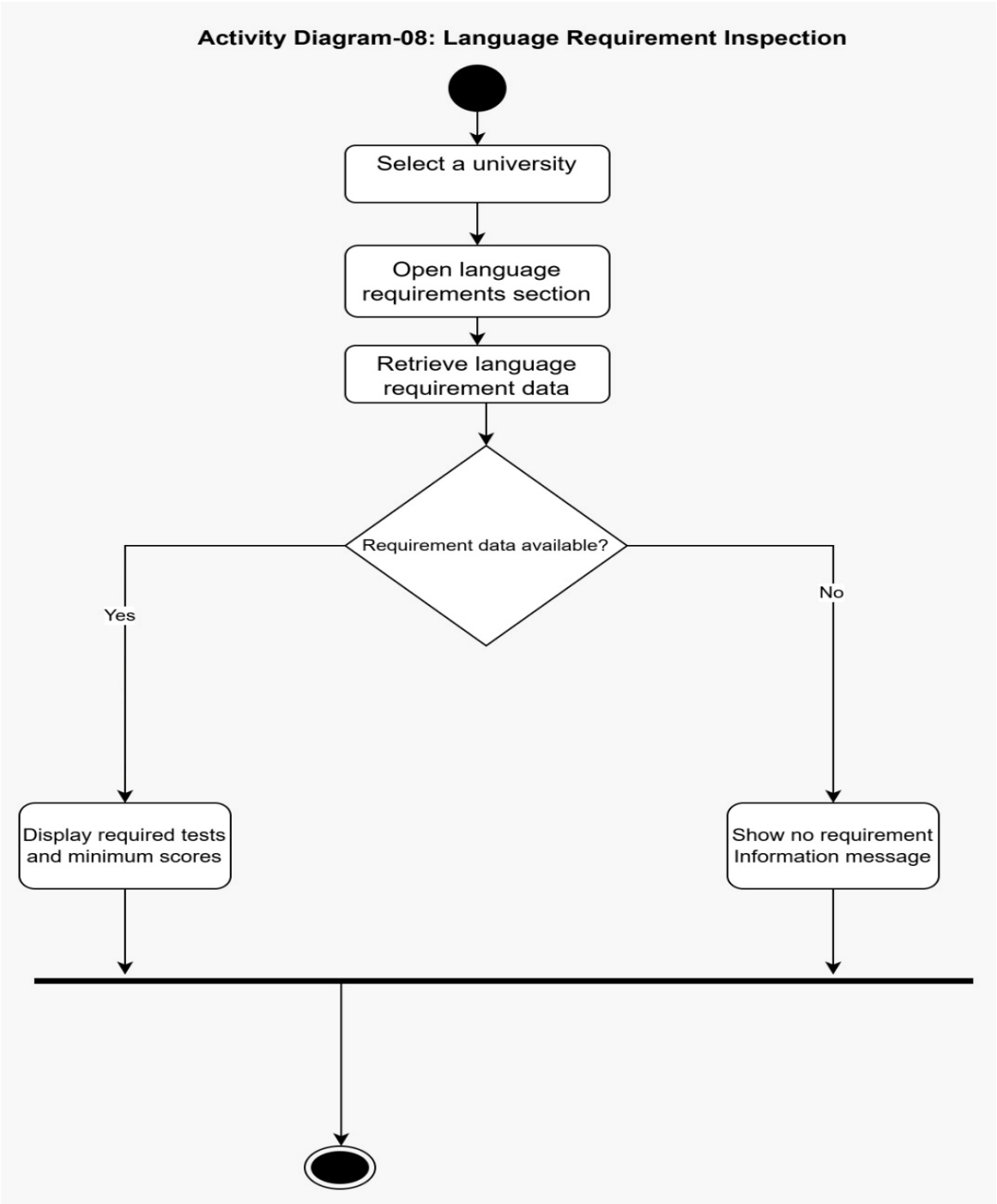
Activity Diagram-06: Academic Program Exploration



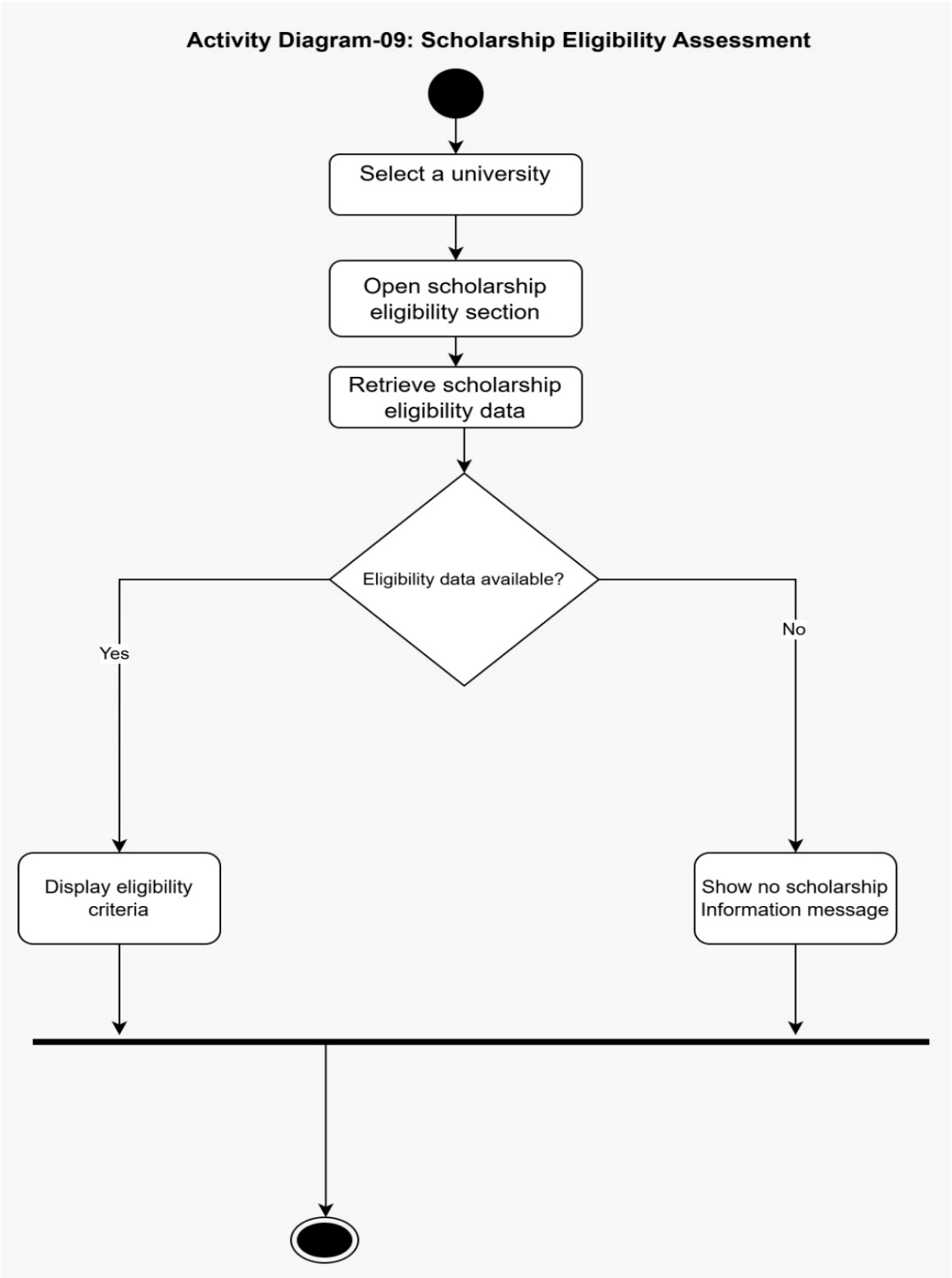
Activity Diagram-07: View Intake Schedule



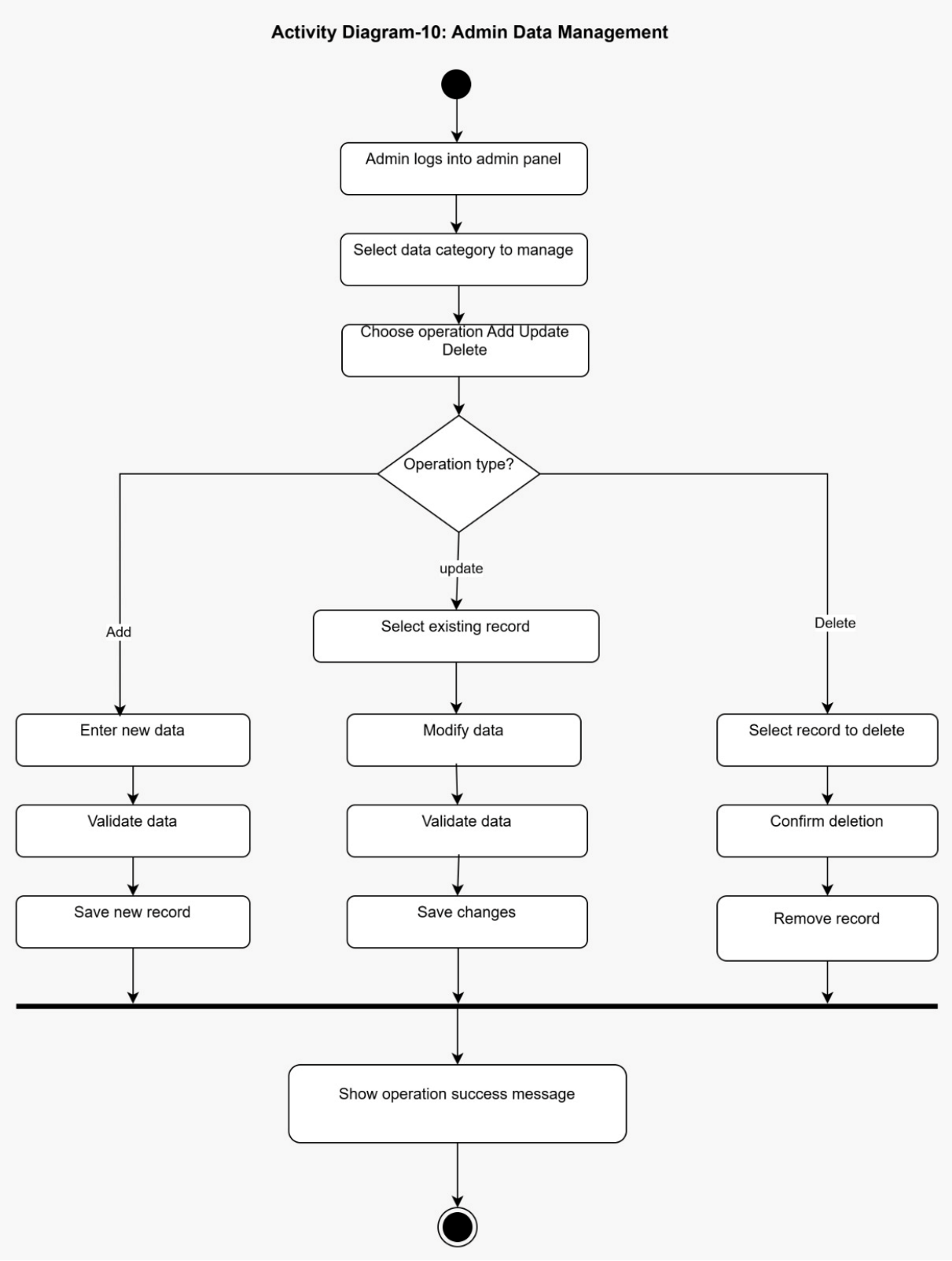
Activity Diagram-08: Language Requirement Inspection



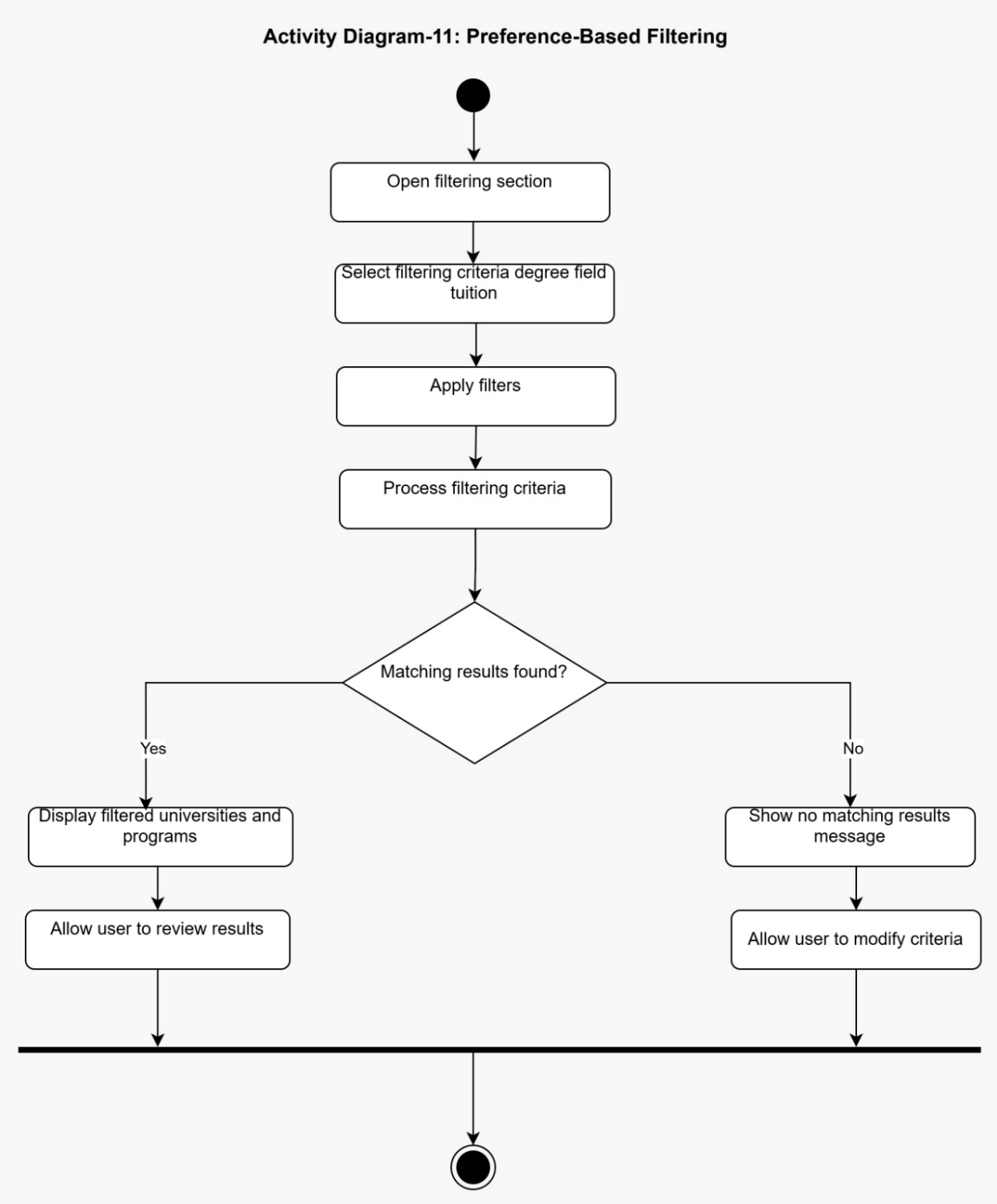
Activity Diagram-09: Scholarship Eligibility Assessment



Activity Diagram-10: Admin Data Management

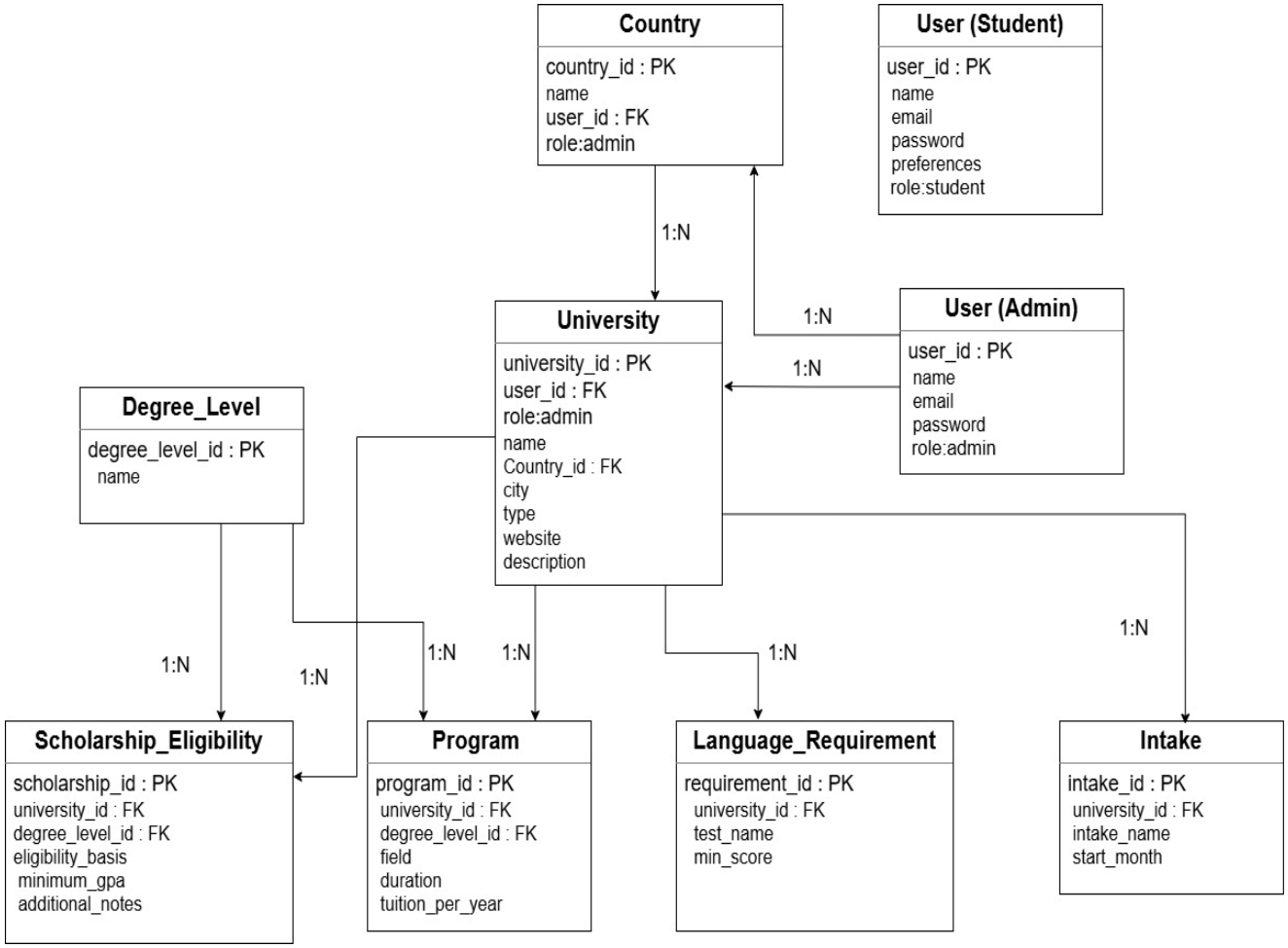


Activity Diagram-11: Preference-Based Filtering



6.4 Entity Relationship Diagram (ERD)

Study Abroad Preparation System ER Diagram



7. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Md. Fazly Rabby	0242310005341077	Frontend development, UI/UX design
Md. Munjurul Islam	0242310005341104	Backend development, Data integration
Kawshik Kumar Saha	0242310005341289	Database design, API implementation, Documentation preparation, Testing and Validation, Backend Design